GROUP WORK PROJECT # ___ MScFE 622: Stochastic Modeling
GROUP NUMBER: _____

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Step 1(a): Heston Calibration using the Lewis Approach

Objective

To calibrate the classical Heston model without jumps to market prices of SM call and put options.

Market Data Used for Step 1(a):

For this part, the 15-day short maturity option prices are used. It contains 5 strikes +1 call & 1 put price at each strike

Current stock price of SM is S0 232.90

Risk Free Rate r = 1.50%

Strike (K)	Market Call Price	Market Put Price
227.5	10.52	4.32
230	10.05	5.2
232.5	7.75	6.45
235	6.01	7.56
237.5	4.75	8.78

The maturity used in model is : T = 15/250=0.06 [250 trading days in one calendar year]

- Both calls and puts are included in calibration.
- Calls are priced directly using Lewis formula.
- Puts are priced using put-call parity .
- Put call parity formula $P = C - S_0 + Ke^{-rT}$
    - P = Put price
    - C = Call price
    - S0 = current price
    - K = Strike Price
    - -r = risk free rate
    - T = Time to maturity

Heston Model Specification

- The Heston model assumes that volatility is not constant and it changes over time
- Variance follows stochastic processes.
- Market prices usually show vol skew or smile effect which is constant vol model does not capture very well.
- The Heston Model can capture this better than model with constant volatility

Under risk –neutral measure- Heston model is

$$dS_t = rS_t dt + sqrt(v_t)S_t dW_t^S$$

And

$$dv_t = kappa(theta - v_t)dt + sigma_v sqrt(v_t)dW_t^v$$

With correlation

$$dW_t^S \cdot dW_t^v = rhodt$$

The parameters calibrated are:

Parameter	Meaning
κ (kappa)	Speed of mean reversion
θ (theta)	Long-run variance
σ_v (sigma_v)	Volatility of variance
ρ (rho)	Correlation between stock and variance shocks
v_o (v0)	Initial variance

In practice, **v_o, ρ, and σ_v** are especially important for short-dated options because these options are very sensitive to the current volatility level and short-term skew.

<u>Pricing Method</u>

We used Lewis (2001) approach to calculate call option prices

Lewis Pricing Approach

The Lewis approach prices option using characteristic function of Heston Model

$$C(S_0, K, T) = S_0 - \frac{\sqrt{S_0 K} e^{-rT}}{\pi} \int_0^\infty \mathrm{Re}\left[\frac{e^{iu\log(S_0/K)}\varphi(u - i/2)}{u^2 + 1/4}\right] du.$$

Here,$\varphi(.)$ is the Heston characteristic function. The integral was evaluated numerically for each strike.

For put options, the model put price was obtained using put-call parity:

$$P = C - S_0 + Ke^{-rT}$$

This is preferred approach as once call price is obtained the put price follows directly from no-arbitrage parity

Error Function Used

- The model is calibrated by minimizing the regular mean squared error or MSE,
- Error function was
- $MSE = \frac{1}{N_C+N_P}\left[\sum_{i=1}^{N_C}\left(C_i^{model} - C_i^{market}\right)^2 + \sum_{j=1}^{N_P}\left(P_j^{model} - P_j^{market}\right)^2\right]$

The parameter are constrained to reasonable range. Variance are kept positive & correlation are restricted to [-1,1].

Put call parity diagnostic

We check the put –call parity condition before moving to calibration .

Strike	Call – Put	$S_0 - K \cdot e^{-rT}$	Parity Deviation
227.5	6.2	5.604658	0.595342
230	4.85	3.106907	1.743093
232.5	1.3	0.609156	0.690844
235	-1.55	-1.888595	0.338595
237.5	-4.03	-4.386346	0.356346

- Table above shows 15- day market price are not perfectly parity consistent.
- Market calls and puts do not satisfy parity exactly, model cannot fit both call and put perfectly at same strike.
- The parity-implied bound MSE is
    - Parity-constrained lower bounds MSE: **0.205585**
- This is the minimum error caused by parity inconsistency alone. The final Heston calibration error should therefore be judged realized to this reasonable bound & not 0

Calibration Process

The calibration process was done as below:

1. SM option with 15 trading days to maturity is selected

2. Use current stock price of S0=232.90
3. Converted maturity to years using T=15/250 = 0.06
4. Used annual risk free rate of 1.5%
5. Priced call options using Lewis formula
6. Calculated put options prices using Put-Call parity
7. Compared model prices with market prices
8. Minimized MSE to find best Heston parameters.
9. Calibration was done using both calls and puts together

Heston-Lewis calibration Results

Parameter	Symbol	Value
Mean reversion speed	κ	3.0225
Long-run variance	θ	0.3197
Volatility of variance	σ_v	1.3167
Correlation	ρ	-0.999
Initial variance	v_o	0.0811
Mean Squared Error	—	0.2464
Root MSE	—	0.4964

- Final MSE is **0.2464** and **final RMSE is 0.4964**
- This means the average pricing error is about 0.50 dollars.

Heston-Lewis Model Fit: Calls

Strike (K)	Market Call	Model Call	Call Error
227.5	10.52	10.5051	-0.0149
230	10.05	8.8523	-1.1977
232.5	7.75	7.3166	-0.4334
235	6.01	5.911	-0.099
237.5	4.75	4.6448	-0.1052

- The model fits some call prices very well . For eg. Strike of 235 call option exactly matches
- Biggest call error is at strike 230.0. The model price is lower than market price by about 1.19

Heston-Lewis Model Fit: Puts

Strike (K)	Market Put	Model Put	Put Error
227.5	4.32	4.9004	0.5804
230	5.2	5.7454	0.5454
232.5	6.45	6.7074	0.2574
235	7.56	7.7996	0.2396
237.5	8.78	9.0312	0.2512

- For put option, model prices are mostly higher than market prices. **This is partly because the model uses put-call parity but the market call & put prices are not perfectly consistent with put call parity.**

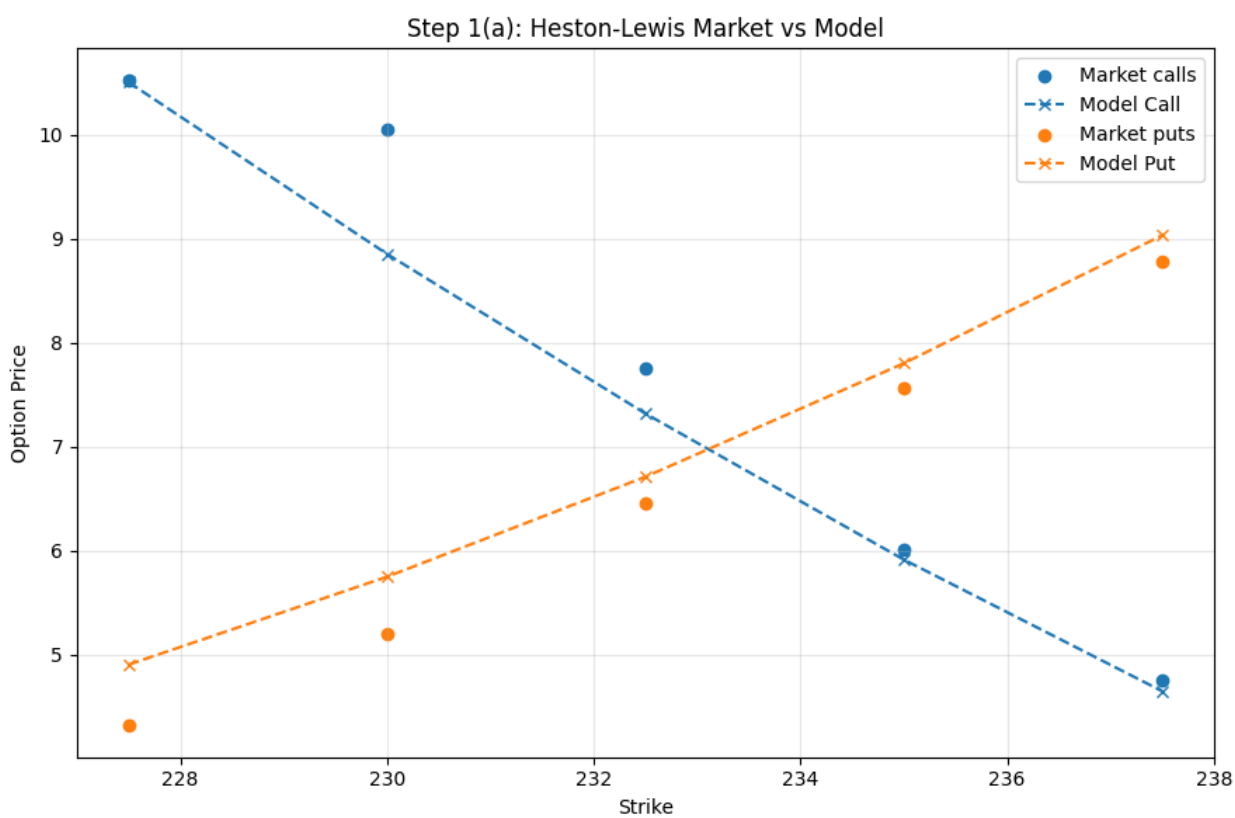

Fig1 : Heston-Lewis calibration fit for 15 day SM options

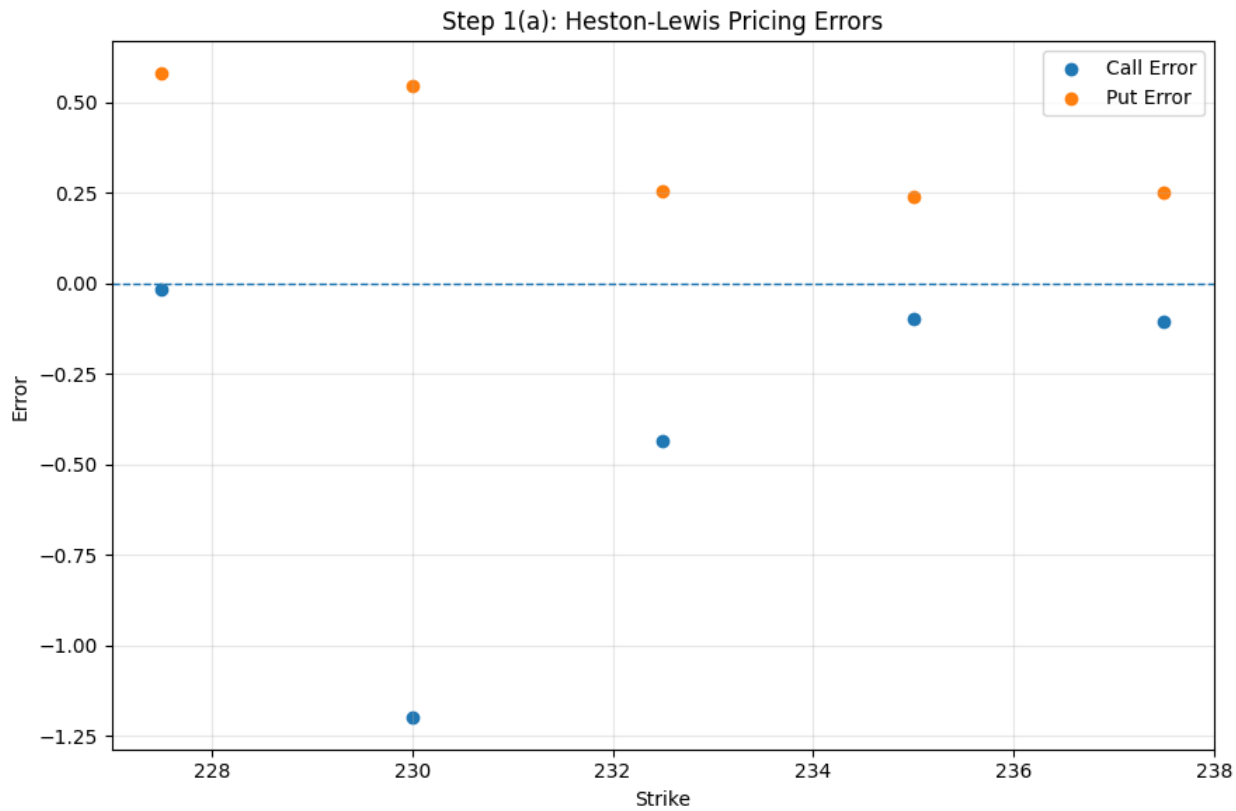

Fig2 : Heston-Lewis pricing error for 15 day SM options

<u>Major observations of Calibration result</u>

- The Heston Lewis calibration gives MSE of 0.2464 and RMSE of 0.4964.
- The fit is reasonable small when compared with parity implied lower bound MSE of 0.205585 .
- The implied larger part of remaining error explained by inconsistencies between market call & put prices, not because of model weakness.
- The calibrated initial variance is v_0 of 0.0811 . This is important as option have only 15 days to maturity
- The correlation parameter is rho= -0.999. This strong negative value indicates stock prices and volatility often moves in opposite direction. When stock price fall, volatility usually increases
- The volatility of variance is sigma_v is 1.316672. This is high meaning model needs variance to move quickly in order to fit short term option prices.
- One limitation is that we have used only 5 stricke and one maturity . With more maturities and more strikes, the calibration would probably be more stable.

Step 2(a): Bates Calibration using the Carr-Madan Approach

Objective:

The new maturity is 60 trading days. For this task –Bates model which is Heston stochastic model with jumps is used as it helps caputre sudden price movement .

The 60 day maturity in annual units is T=60/250=0.24

Current SM stock price S0= 232.90

Risk free rate = 1.50%

Market Data used for Step 2(b)

For 60 day maturity, the option price data used for this calibration is

Strike (K)	Market Call Price	Market Put Price
227.5	16.78	11.03
230	17.65	12.15
232.5	16.86	13.37
235	16.05	14.75
237.5	15.1	15.62

- Both call and put prices are used
- Call price are calculated using Carr-Madan pricing approach
- Put prices are calculated using put call parity: $P = C - S_0 + Ke^{-rT}$

Bates Model Specification

- Bates model is an extension of the Heston model.
- The Heston model already allow volatility to change over time. Bates adds jumps feature in stock price

The stock-price process can be written as:

$$\frac{dS_t}{S_t} = (r - \lambda\kappa_J)dt + \sqrt{v_t}dW_t^S + (J-1)dN_t.$$

Variance process:

$$dv_t = \kappa(\theta - v_t)dt + \sigma_v\sqrt{v_t}dW_t^v.$$

Jump compensation is

$$\kappa_J = E[J-1] = e^{\mu_J + \frac{1}{2}\delta_J^2} - 1.$$

This compensation term is included so that the model is risk-neutral.

Bates Model Parameters calibrated in this model

Parameter	Meaning
κ (kappa)	Speed of variance mean reversion
θ (theta)	Long-run variance
σ_v (sigma_v)	Volatility of variance
ρ (rho)	Correlation between stock and variance shocks
v_o (v0)	Initial variance
λ_j (lambda_j)	Jump intensity
μ_j (mu_j)	Mean log-jump size
δ_j (delta_j)	Jump-size volatility

The extra jump parameters help to model sudden stock price movements

<u>Carr-Madan Pricing Method</u>

- The Carr–Madan method is used to calculate prices for call option prices.
- The Carr-Madan method uses characteristic function of Bates model.
- It applies Fourier transform to calculate option prices .
- For log strike k=log K = \log Kk=logK and damping parameter α>0\alpha > 0α>0, the call price is:
- Carr-Madan call pricing formula is

$$C(K) = \frac{e^{-\alpha k}}{\pi}\int_0^\infty \mathrm{Re}\left[e^{-iuk}\frac{e^{-rT}\varphi\big(u - i(\alpha+1)\big)}{\alpha^2 + \alpha - u^2 + i(2\alpha+1)u}\right]du$$

- _k = log K
- $\varphi(.)$ is the Bates characteristic function
- Damping parameter uses are
  - alpha=1.5
- After each call price is calculated, the corresponding put price is calculated using put-call parity $P = C - S0 + Ke - rT$
- This allows calibration to use both call & put prices

Calibration Error Function Used

- Calibration was done by minimizing regular mean squared error or MSE
- Error function is

$$MSE = \frac{1}{N_C + N_P}\left[\sum_{i=1}^{N_C}\left(C_i^{model} - C_i^{market}\right)^2 + \sum_{j=1}^{N_P}\left(P_j^{model} - P_j^{market}\right)^2\right]$$

- Nc & Np are number of call & put option respectively.
- Model tried to find Bates parameters that made model option prices as close to possible market prices.

Calibration Process:

- SM option with 60 trading days are selected
- SM current stock price 232.90 , T=0.24 r= 0.015
- Bates model is used
- Call option are prices using Carr-Madan method
- Put price calculated using put-call parity
- Minimized regular MSE to find best Bates model parameters

Put Call Parity Diagnostic

Strike (K)	Call – Put	$S_0 - K \cdot e^{-rT}$	Parity Deviation
227.5	5.75	6.2175	-0.4675
230.0	5.50	3.7265	1.7735
232.5	3.49	1.2355	2.2545
235.0	1.30	-1.2555	2.5555
237.5	-0.52	-3.7465	3.2265

Parity-constrained lower bounds MSE: **1.269393**

Bates-Carr-Madan calibration Results

Parameter	Symbol	Value
Mean reversion speed	κ	0.01
Long-run variance	θ	0.000001
Volatility of variance	σ_v	4.3824
Correlation	ρ	-0.999
Initial variance	v_0	0.0266
Jump intensity	λ_j	1.9018

Jump mean	μ_j	0.2074
Jump volatility	δ_j	0.0262
Mean Squared Error	—	1.3268
Root MSE	—	1.1518

- Final **MSE is 1.326750** and **RMSE is 1.151847**
- **The** average pricing error is about 1.15 dollars

Bates-Carr-Madan Model Fit- Calls

Strike (K)	Market Call	Model Call	Call Error
227.5	16.78	17.3340	0.5540
230.0	17.65	16.4681	-1.1819
232.5	16.86	15.5792	-1.2808
235.0	16.05	14.6549	-1.3951
237.5	15.10	13.7312	-1.3688

Bates-Carr-Madan Model Fit- Puts

Strike (K)	Market Put	Model Put	Put Error
227.5	11.03	11.1165	0.0865
230.0	12.15	12.7416	0.5916
232.5	13.37	14.3437	0.9737
235.0	14.75	15.9104	1.1604
237.5	15.62	17.4777	1.8577

- The model fits lower strike well. For eg strick 227.50 model price is 17.33 while market call is 16.78
- For higher strickes the model generaly underprices calls and overprices the puts. This connected to put-call parity issue in market data

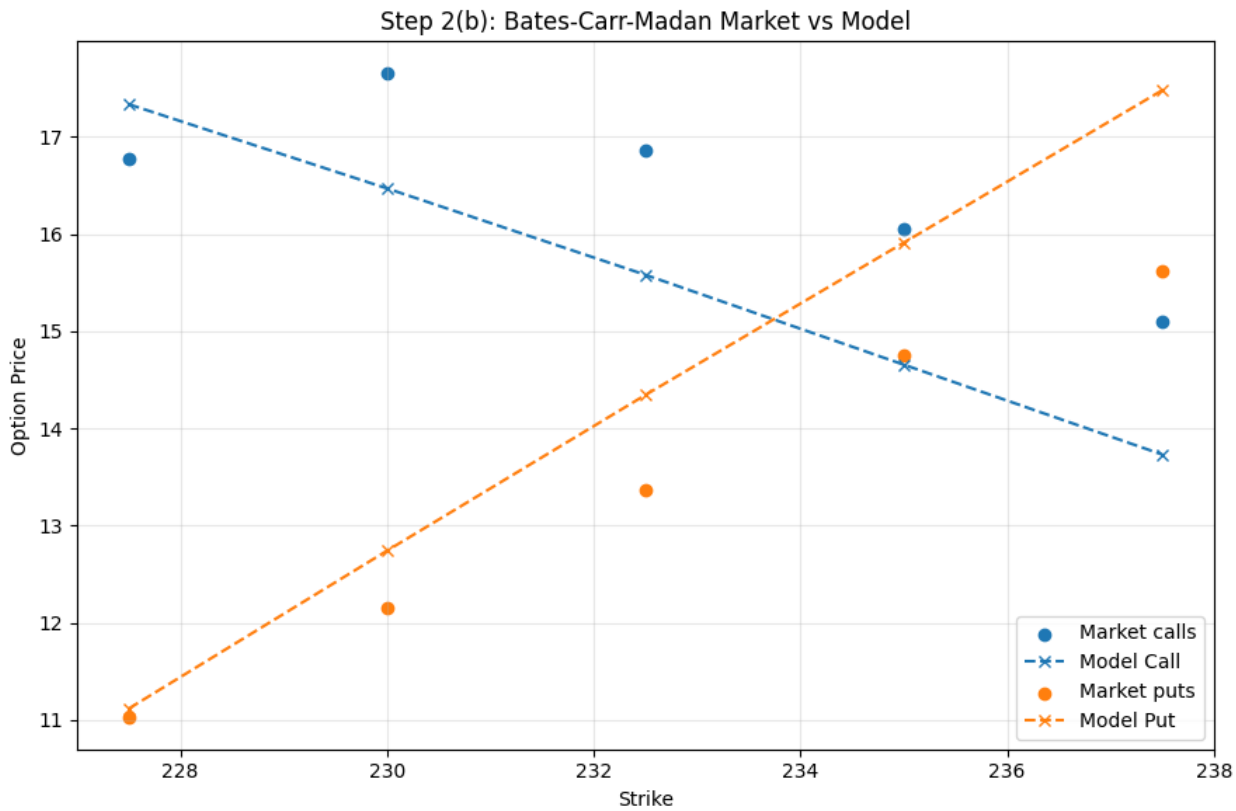

Fig 3: Bates-Carr Madan calibration fit for 60 days SM options

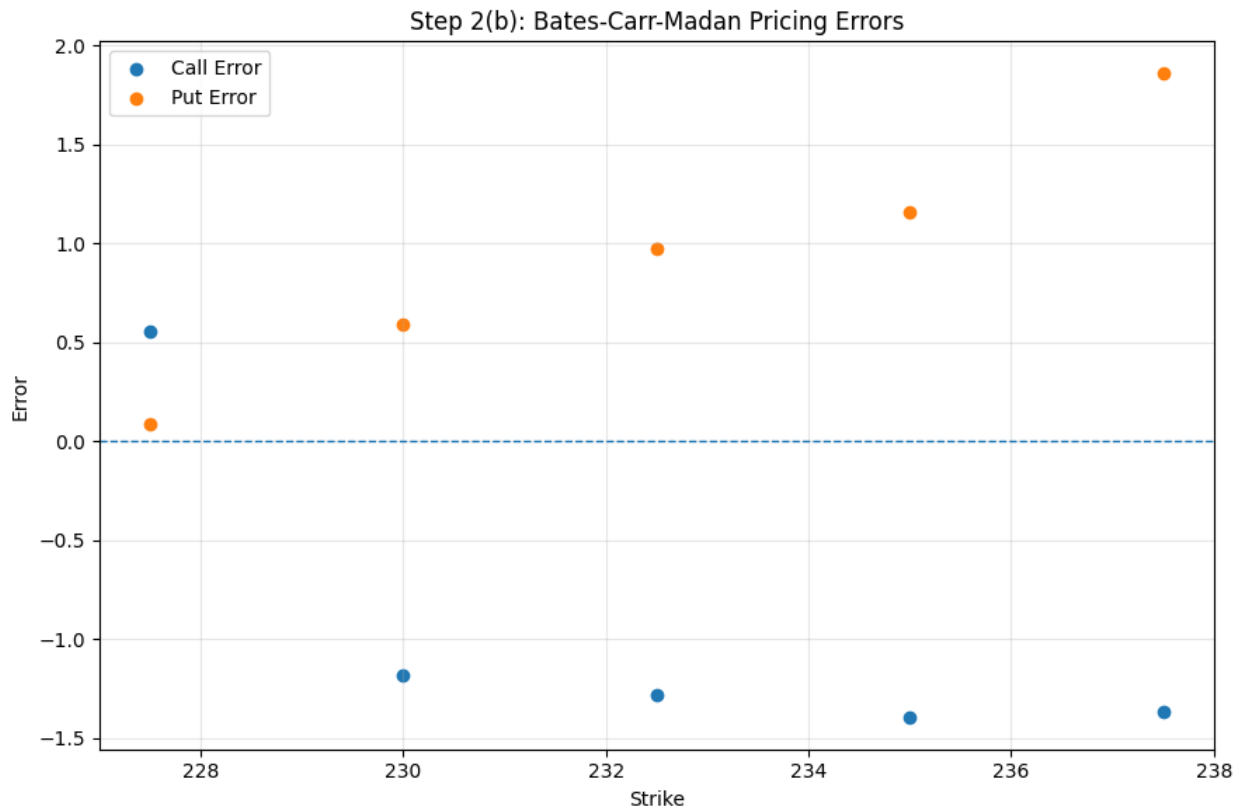

Fig 4: Bates-Carr Madan pricing errors for 60 days SM options

<u>Major observations of Bates calibration results:</u>

- Final MSE is 1.32675 is higher than Step 1(a) Heston calibration. This is because 60 days market has larger differences between call & put prices when checked against put-call parity.
- The jump intensity is 1.901840 suggesting the jump risk is important to 60 day maturity
- Average jump size is 0.207446 is positive meantinge positive jump on average.
- Jump volatility is 0.026247 suggesting jump size do not vary too much in calibration
- Some parameters are close to their bounds suggesting the calibration is not perfectly stable.
    - K=0.01
    - Rho=-0.99
- This can happen because we have only one maturity and 5 stricke.
- Bates model has many parameters so it can be hard to estimate all of them accurately from small dataset